Title: A Motion-Robust Dual-Echo 3D LGE Reconstruction Framework

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Background: CMR-based late gadolinium enhancement (LGE) provides high-resolution myocardial scar visualization and fibrosis quantification, enabling risk stratification and management across cardiomyopathies.[1,2] While conventional 2D LGE is hindered by multiple breath-holds, limited coverage, and poor through-plane resolution, free-breathing whole-heart LGE delivers isotropic coverage, enhancing scar detection and patient comfort.[3,4] State-of-the-art 3D implementations rely on image navigation (iNAV) to either align images across heartbeats[5] or perform respiratory bining[6] to improve data efficiency from a ~5 to 10 minutes scan. Nonetheless, low signal-to-noise ratio and residual motion blurring remain a challenge in 3D LGE. Moreover, these motion-correction or compensation techniques cannot accommodate patients with frequent arrhythmia. By extending our recently proposed highly accelerated real-time cine ML-DIP framework,[7] we propose a motion-robust reconstruction framework for 3D LGE, called multi-dynamic low-rank multi-encoding deep image prior with sensitivity maps estimation (MLE-DIPS).

Methods: We extended ML-DIP to 3D LGE by jointly estimating sensitivity maps and enabling dual-echo reconstruction from a shared spatial basis to exploit inter-echo redundancy (Fig. 1). We validated the resulting MLE-DIPS in five ventilated Yucatan minipigs and five human subjects scanned on a 3T scanner (MAGNETOM Vida, Siemens Healthineers) using a ~5-minute single-shot free-breathing dual-echo iNAV sequence.[5] In minipigs, rupture of the mitral valve chordae tendineae induced mitral regurgitation with left atrial enlargement. Human patients were randomly selected from those clinically referred for CMR-LGE. Imaging parameters for both studies included spatial resolution of 1.0×1.0×1.25 to 1.25×1.25×1.30 mm^3, temporal footprint of 180–240 ms, and per-frame acceleration rates of 768–1672. The coil data were compressed to six virtual channels using ROVir[8] to enhance the signal from the cardiac region. Reconstructions were compared against an iNAV-based inline reconstruction[5] and LR-DIP[9] extended to 3D (Fig. 2, Table 1). Fat-water separation was performed post-reconstruction using IDEAL-CE.[10] For all subjects, a CMR expert scored each method (3D movies and static slices) on a five-point Likert scale (1: non-diagnostic, 5: excellent).

Results: In both animal and human studies (Fig. 2), iNAV reconstructions appeared noisy and LR-DIP overly smooth, whereas MLE-DIPS effectively suppressed motion artifacts

while preserving detail. Quantitative scores (Table 1) showed a consistent advantage for MLE-DIPS over iNAV.

Conclusion: We introduce MLE-DIPS, a self-supervised deep-learning reconstruction framework for motion-robust 3D LGE, demonstrating feasibility in animal and human studies using a ~5-minute free-breathing acquisition. Future work will integrate Dixon modeling in MLE-DIPS for improved image synthesis, robust fat-water separation, and more confident scar-fat differentiation, with validation in a larger cohort.

References:

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Figures:

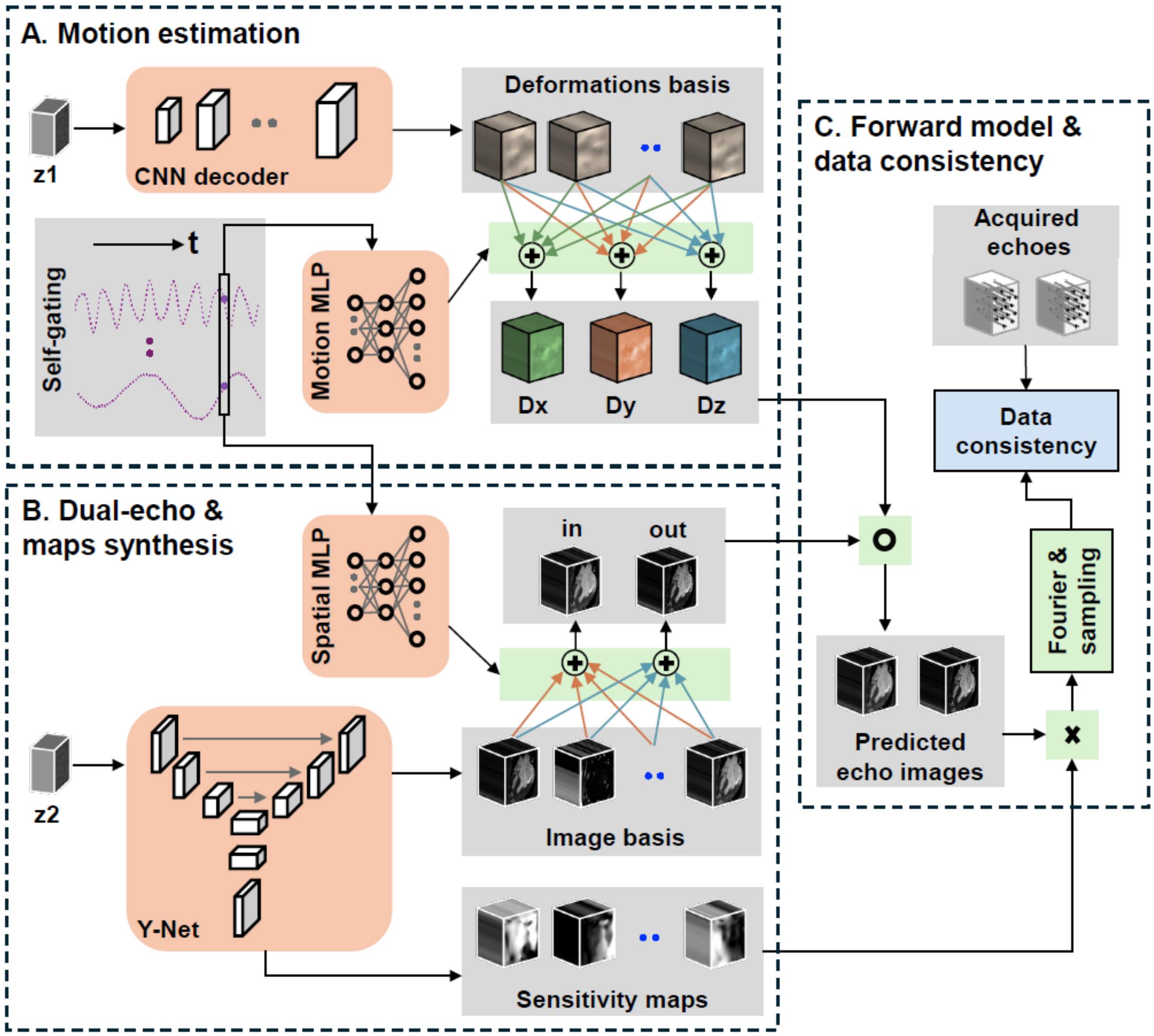

Fig. 1: An overview of the proposed MLE-DIPS framework. (A) Motion estimation: A CNN decoder maps a static code vector z1 to a 3D deformation basis consisting of building blocks, which are aggregated using time-varying weights generated by a motion MLP driven by trainable dynamic code vectors (or self-gating) to form three deformations corresponding to three spatial directions, represented by Dx, Dy, and Dz. (B) Dual-echo images and sensitivity maps synthesis: With input z2, a Y-Net generates spatial basis as well as smooth sensitivity maps. In addition, a spatial MLP generated echo- and time-specific weights to synthesize in-phase and out-of-phase images from the shared spatial basis. (C) Forward model and data consistency: The echo images from (B) are spatially warped ("**o**") with estimated motion fields from (A), multiplied ("**x**") with generated sensitivity maps from (B), applied Fourier and sampling operations, and finally made consistent with acquired data. This joint formulation avoids data binning/rejection and can potentially remain robust in arrhythmia.

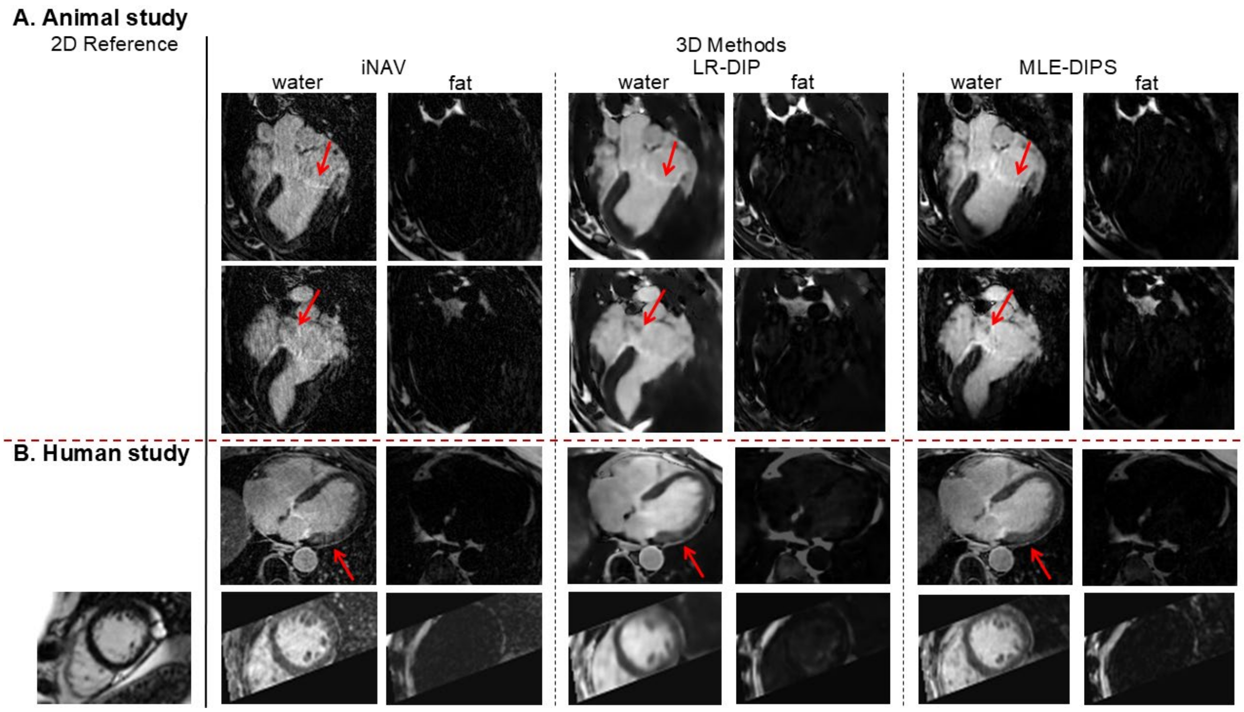

Fig. 2: Qualitative comparison in animal and human studies. (A) Two representative slices from a Yucatan minipig. (B) A representative slice from a human subject: the first row shows a selected slice, while the second row includes a slice matching 2D SAX reference. Each two-column subgroup corresponds to a competing 3D reconstruction method (iNAV, LR-DIP, and MLE-DIPS), showing both water and fat images. iNAV images look sharp but suffer from noise amplification, LR-DIP blurs details, and MLE-DIPS effectively suppresses noise while preserving sharp myocardial detail, as highlighted by red arrows.

Table:

	iNAV	LR-DIP	MLE-DIPS
Pigs (5)	4.2	3.4	**4.4**
Humans (5)	3.8	2.8	**4**
Both (10)	4	3.1	**4.2**

Table 1: CMR expert scoring of reconstructions in terms of overall image quality in animal and human studies. The first row reports mean image-quality scores across five pigs, and the second row across five human subjects, each reviewed in a blinded manner. The third row shows combined averages over all ten subjects. Bold values indicate the best score.